

Problem 15.7

Show that

$$s_{\overline{m+n}|} = s_{\overline{m}|} + (1+i)^m s_{\overline{n}|} = s_{\overline{n}|} + (1+i)^n s_{\overline{m}|}.$$

Interpret this result verbally.

Solution

Recall that the accumulated value of an annuity-immediate with payments of 1 is

$$s_{\overline{k}|} = \frac{(1+i)^k - 1}{i}.$$

First, consider

$$s_{\overline{m}|} + (1+i)^m s_{\overline{n}|}.$$

Using the formula for $s_{\overline{k}|}$, we get

$$s_{\overline{m}|} + (1+i)^m s_{\overline{n}|} = \frac{(1+i)^m - 1}{i} + (1+i)^m \left(\frac{(1+i)^n - 1}{i} \right).$$

Combining terms,

$$= \frac{(1+i)^m - 1 + (1+i)^m ((1+i)^n - 1)}{i}.$$

Expanding the numerator,

$$= \frac{(1+i)^m - 1 + (1+i)^{m+n} - (1+i)^m}{i}.$$

The terms $(1+i)^m$ cancel, so

$$= \frac{(1+i)^{m+n} - 1}{i}.$$

Therefore,

$$s_{\overline{m}|} + (1+i)^m s_{\overline{n}|} = s_{\overline{m+n}|}.$$

Similarly, consider

$$s_{\overline{n}|} + (1+i)^n s_{\overline{m}|}.$$

Using the same formula,

$$s_{\overline{n}|} + (1+i)^n s_{\overline{m}|} = \frac{(1+i)^n - 1}{i} + (1+i)^n \left(\frac{(1+i)^m - 1}{i} \right).$$

Combining terms,

$$= \frac{(1+i)^n - 1 + (1+i)^n ((1+i)^m - 1)}{i}.$$

Expanding the numerator,

$$= \frac{(1+i)^n - 1 + (1+i)^{m+n} - (1+i)^n}{i}.$$

The terms $(1+i)^n$ cancel, so

$$= \frac{(1+i)^{m+n} - 1}{i} = s_{\overline{m+n}|}.$$

Thus,

$$\boxed{s_{\overline{m+n}|} = s_{\overline{m}|} + (1+i)^m s_{\overline{n}|} = s_{\overline{n}|} + (1+i)^n s_{\overline{m}|}}.$$

Verbal Interpretation

An annuity with $m+n$ payments can be split into two blocks. One block contains m payments and the other contains n payments. If the first block is valued at time $m+n$, it must be accumulated forward to that time, while the second block is already accumulated over its own term. The same total value is obtained no matter which block is considered first.